

EX.NO : 14

IMPLEMENTATION OF SELECTIVE REPEAT PROTOCOL

AIM :

To write a Java program to perform a Selective Repeat Protocol.

ALGORITHM :

1. Start program, prompt frame size.
2. Transmit frames, prompt lost Acknowledgement.
3. Print acknowledgment or retransmission message based on lost Acknowledgement.
4. Repeat until lost Acknowledgement equals frame size.
5. Stop program.

PROGRAM :

```
import java.io.*;

public class SelectiveRepeat
{
    public static void main(String args[]) throws IOException
    {
        BufferedReader br = new BufferedReader(new
        InputStreamReader(System.in));

        System.out.println("Enter the Frame Size: ");

        int window = Integer.parseInt(br.readLine());

        for (int sent = 0; sent < window; sent++)
        {
```

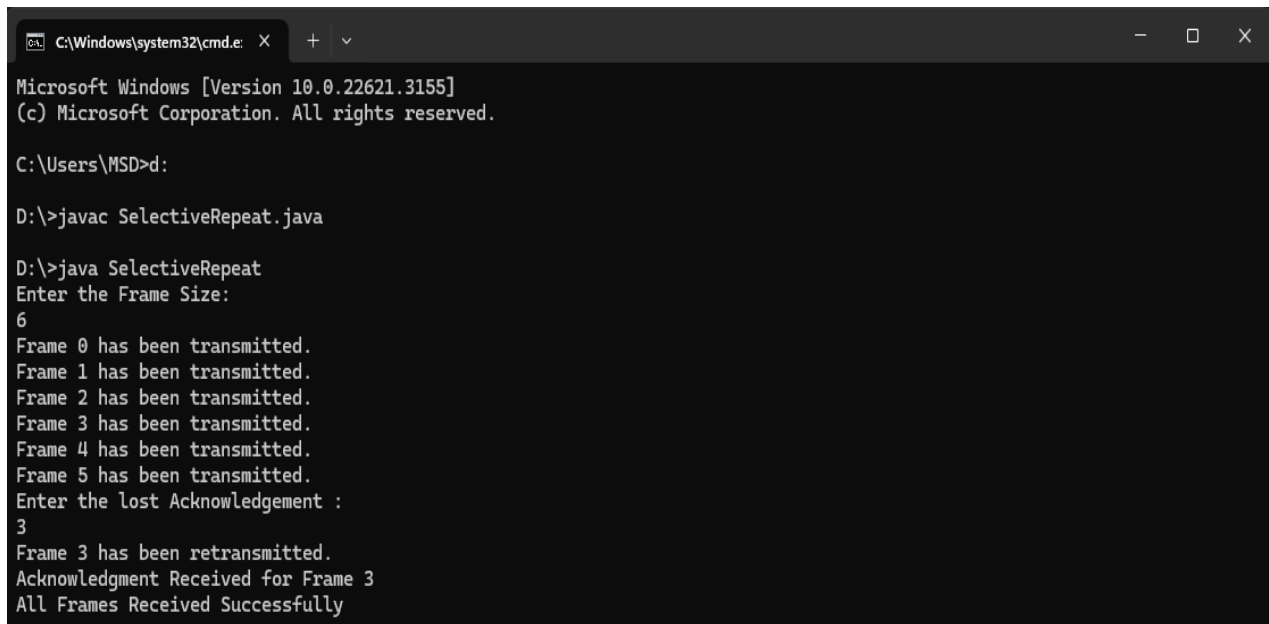
```

        System.out.println("Frame " + sent + " has been transmitted.");
    }
    int ack;
    do
    {
        System.out.println("Enter the lost Acknowledgement : ");
        ack = Integer.parseInt(br.readLine());
        if (ack == window)
        {
            System.out.println("Acknowledgment Received for Frame "
+ (ack - 1));
        }
        else
        {
            System.out.println("Frame " + ack + " has been
retransmitted.");
            System.out.println("Acknowledgment Received for Frame "
+ ack);
            System.out.println("All Frames Received
Successfully");
        }
    }

    while (ack == window);
}

```

OUTPUT :



```
C:\Windows\system32\cmd.exe X + v
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSD>d:

D:\>javac SelectiveRepeat.java

D:\>java SelectiveRepeat
Enter the Frame Size:
6
Frame 0 has been transmitted.
Frame 1 has been transmitted.
Frame 2 has been transmitted.
Frame 3 has been transmitted.
Frame 4 has been transmitted.
Frame 5 has been transmitted.
Enter the lost Acknowledgement :
3
Frame 3 has been retransmitted.
Acknowledgment Received for Frame 3
All Frames Received Successfully
```

RESULT :

Thus, the program for Implementing a Selective Repeat Protocol is successfully executed and the output is verified.